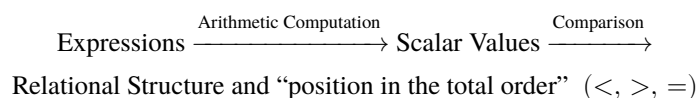


Comparing and Totally Ordering Mathematical Expressions: Are Traditional Processing Methods Still Suitable Today?

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Abstract

I call this the “classic processing workflow”:



Observing the center: the primary reason we use the real numbers $\mathbb{R}$ today lies in their total ordering and the ability to compare them ($<$, $>$, $=$). Is the intermediate step of “inferring a scalar value” a mathematical ontological necessity for transforming an expression into a relational structure and “position in the total order”? Or does an alternative mathematical space exist that allows extracting such information directly from the structure of the expression, thereby bypassing the step of inferring the scalar value—whether partially or completely?

This is not a proposal to eliminate computation. The goal is to redefine the concept of computation without necessarily relying on intermediate scalar values, thereby yielding benefits.

Contents

1	Introduction	1
2	Definition	2
3	Two Major Issues	2
4	Definition of COST_ALL, Classical Baseline	3
5	An Open Problem or an Impossible One?	4
6	Avenues of Attack	4
6.1	Tropical Geometry	4
6.2	Operator Algebras and Hilbert Spaces — For Problem 2 (Inner Product)	5
	References	6
	Index	7

1 Introduction

For millennia, the practice of mathematics and computation has rested on a single, largely unquestioned pipeline: an expression is evaluated into a scalar—a point on the real line $\mathbb{R}$ —and only then can it be compared against another such point. Addition, subtraction, multiplication, and division are treated as operations whose sole legitimate output is a number; ordering ($<$, $>$, $=$) is treated as a relation that can only be asked after that number has been produced. This paper calls that pipeline the *classic processing*

workflow and asks a deliberately naive-sounding question: is the intermediate scalar actually necessary, or is it merely the historically dominant—and possibly wasteful—route to the answer we actually want, namely an expression’s position in a total order?

Section 2 formalizes this question as three requirements (A1–A3) that any candidate space $\mathbb{S}$ must satisfy to be considered a genuine alternative to $\mathbb{R}$ for this purpose. Section 3 splits the inquiry into two concrete open problems: the fully general case of arithmetic expressions over $\{+, -, \times, \div\}$, and the structurally constrained case of the inner (scalar) product. Section 4 defines the cost model, COST_ALL , that gives the word “cheaper” an operational, hardware-grounded meaning, together with the classical baseline against which any candidate $\mathbb{S}$ must be measured. Section 5 surveys the current state of knowledge, arguing that the problem is open rather than settled, with partial precedents on both sides. Section 6 sketches two candidate avenues of attack—tropical geometry and operator-algebraic/Hilbert-space methods—as illustrations of how a structural, non-scalar semantics for ordering might be built, without claiming that either avenue currently constitutes a complete solution.

2 Definition

Requirement 1 (Comparability). *There exists a relation operator $\leq_{\mathbb{S}}$, defined directly on the structure of two elements $A, B \in \mathbb{S}$, such that:*

$$\leq_{\mathbb{S}}(A, B) \iff \text{eval}(A) \leq \text{eval}(B).$$

In other words, $\leq_{\mathbb{S}}$ is logically equivalent to the standard ordering relations $\{<, >, =\}$ on scalar values—but the left-hand side does not involve the explicit, full computation of $\text{eval}(A)$ and $\text{eval}(B)$.

Requirement 2 (Structural Distance / Distance Metric). *There exists a function*

$$d_{\mathbb{S}} : \mathbb{S} \times \mathbb{S} \rightarrow \mathbb{R}_{\geq 0}$$

(or an equivalent ordering structure), defined directly on the expressions, such that $d_{\mathbb{S}}(A, B)$ accurately reflects the difference $|\text{eval}(A) - \text{eval}(B)|$ (or a monotonic transformation thereof) without requiring the full computation of $\text{eval}(A)$ and $\text{eval}(B)$.

Requirement 3 (Total Ordering: Strongly Equivalent to $\mathbb{R}$, but not $\mathbb{R}$). *(Read the definition of COST_ALL below.)*

$\mathbb{S}$ must support a total ordering relation with expressive power equivalent to that of $\mathbb{R}$ (sufficient to distinguish between any pair of distinct finite-precision values as required by the problem).

To clarify the constraint of “avoiding issues associated with $\mathbb{R}$ ”: this is not a constraint regarding the nature of the set itself (such as cardinality, countability, or the finite representability of $\mathbb{R}$ on computers—an issue effectively resolved in practice by finite-precision floating-point arithmetic). The actual constraint lies in the processing cost— COST_ALL (defined in Section 4): if $\mathbb{S}$ were simply forced to reduce to $\mathbb{R}$ before a comparison could be made, it would revert exactly to the “classical processing workflow” that the problem aims to avoid, rendering the initial objective cost-ineffective—not “set-theoretically impossible,” but rather “cost-wise trivial or futile.”

If it is simply “isomorphic” to $\mathbb{R}$ yet requires a lower COST_ALL , that is impossible.

Therefore, A3 is understood as follows: $\mathbb{S}$ must possess a total order capable of discrimination equivalent to that of $\mathbb{R}$, while the information extraction process must incur a COST_ALL that is genuinely lower than that of the “reduce to $\mathbb{R}$ then compare” approach.

3 Two Major Issues

Main Problem 1 (General Case). **Given:** N expressions, each containing only basic arithmetic operations $\{+, -, \times, \div\}$ with operands being real numbers of finite precision.

Exclusion: All transcendental functions. According to Richardson’s Theorem [1] (1968), determining whether $\text{eval}(A) = 0$ holds is undecidable in the general case when expressions are permitted to contain transcendental functions; thus, this exclusion is a necessary condition for the problem to remain decidable, rather than an arbitrary choice by the author.

Open question: Does there exist a system $\mathbb{S}$ —along with algorithms satisfying requirements A1–A3 for this class of finite-precision expressions $\{+, -, \times, \div\}$ —that achieves a COST_ALL level lower than the baseline?

Main Problem 2 (The Scalar Product Case). **Given:** N expressions of a specific (non-arbitrary) form:

$$\langle u, v \rangle = \sum_{i=1}^n u_i v_i,$$

where u_i and v_i are real numbers with finite precision.

Open question: With stricter structural constraints (a fixed sum-of-products form rather than an arbitrary expression), are problems A1–A3 easier to solve than Main Problem 1? In other words, does this specific sum-of-products structure allow for a feasible $\mathbb{S}$ that might be difficult to find in the general case?

4 Definition of COST_ALL , Classical Baseline

$$\text{COST_ALL} = \text{Cost}_{\text{computation}} + \text{Cost}_{\text{memory/representation}} + \text{Cost}_{\text{etc}}$$

measured in a real-world environment using specific hardware—not in a computational model that abstracts away resources.

A1–A3 must be feasible within a “realistic environment and specific hardware”—meaning they are subject to hardware constraints or specific computational models, rather than operating within an idealized computational model with infinite resources. The “total actual implementation cost” (COST_ALL , Section 3) serves as the metric for feasibility—not merely theoretical feasibility (where existence alone suffices), but practical feasibility (where COST_ALL demonstrates a genuine reduction compared to the baseline).

Specific cost model breakdown

$\text{Cost}_{\text{computation}}$

This represents the actual processing cycles and operations required by the hardware to process the expressions. In traditional methods, this involves executing all basic arithmetic operations $\{+, -, \times, \div\}$ to their full required finite precision.

$\text{Cost}_{\text{memory/representation}}$

This accounts for the memory bandwidth and storage required during processing within a realistic environment. It includes the cost of maintaining and storing intermediate scalar values at finite precision before a final relational comparison can be made.

Cost_{etc}

This encompasses other hardware-specific constraints and overheads inherent to the specific computational model being used, as opposed to an idealized computational model with infinite resources.

Definition of Classical Baseline

The Classical Baseline relies entirely on the “classic processing workflow.” In this workflow, the system must first perform an arithmetic computation to transform expressions into scalar values representing a position on the real number line $\mathbb{R}$. Specifically, it requires the explicit and full computation of $\text{eval}(A)$

and $\text{eval}(B)$. Only after these intermediate scalar values are inferred can the system apply standard ordering relations ($<$, $>$, $=$).

An $\mathbb{S}$ -system is considered ‘excellent’ only if its COST_ALL is significantly reduced.

5 An Open Problem or an Impossible One?

(i) There is no proof of general impossibility. No theorem—including Richardson’s Theorem, which merely excludes transcendental functions from the scope of the problem—demonstrates the non-existence of an $\mathbb{S}$ satisfying A1–A3 with a COST_ALL lower than the baseline. Richardson’s Theorem limits the scope of the problem (excluding transcendental functions to preserve decidability) but does not limit the existence of $\mathbb{S}$ within the remaining scope.

(ii) There is partial precedent—evidence indicating that this non-trivial problem is not impossible:

Regarding Main Issue 1: exact or certified geometric predicate evaluation techniques (such as interval arithmetic and adaptive-precision arithmetic—exemplified by the work of Shewchuk [2] and Yap [3] in computational geometry) satisfy requirement A1 (determining the sign or order of an expression) at a lower average cost than fully reducing the expression to an exact scalar value, although these techniques do not necessarily satisfy all three axioms (A1–A3) in every case.

Regarding Issue 2: techniques such as Locality-Sensitive Hashing (LSH) [4], random projection (the Johnson–Lindenstrauss lemma) [5], and Maximum Inner Product Search (MIPS) [6] achieve A1/A2 approximations for the inner product problem, yielding a COST_ALL lower than the baseline across various real-world data scenarios. However, since these are probabilistic or approximate solutions that do not satisfy A3 (which requires an exact total ordering free of error), the question of whether an exact (non-approximate) solution exists remains open.

Based on points (i)–(ii), the core problem in Section 4 should be classified as an open problem: there is neither a general solution nor a general counterexample or impossibility theorem, yet there are sufficient partial precedents (in exact geometric computation and in sketching/LSH/MIPS) to indicate that this direction is not trivially impossible. The level of difficulty varies between the two sub-cases: Core Problem 2 (inner product) possesses a more tightly constrained structure and already has many efficient approximation solutions, making an exact solution more feasible; conversely, Core Problem 1 (general arithmetic expressions) involves a space of expressions not constrained by any specific form.

6 Avenues of Attack

To satisfy A1–A3, we cannot simply redesign computer algorithms; we must redefine their nature at the level of mathematical ontology. Does the millennia-long reliance on “reducing an expression to a single scalar point on the real number line $\mathbb{R}$ ” represent a waste of structural information?

The core objective of this section is to construct mappings from the Space of Expressions ($\mathbb{E}$) directly to the Space of Order Relations ($\mathbb{O}$), completely bypassing the scalar space $\mathbb{R}$. The following approaches are based on branches of modern abstract mathematics. **In reality, highly innovative efforts may be required.**

6.1 Tropical Geometry

Tropical geometry [7] revolutionized algebra by redefining operations: replacing addition with the minimum or maximum operation, and multiplication with addition (e.g., $x \oplus y = \max(x, y)$). This transforms complex algebraic curves into piecewise-linear polytopal graphs.

Approach: We can construct a form of “Extended Tropical Geometry” to map expressions from Problem 1 and Problem 2 onto geometric structures (polytopes) in a high-dimensional space. The distance

function d_S (Requirement A2) would then correspond to the Hausdorff distance or Wasserstein distance between these two polyhedra. Comparing two polyhedra in geometry can sometimes be achieved by considering only their boundary vertices or faces—using measure theory—while completely disregarding their “interior” (which corresponds to intermediate calculations). This approach is particularly feasible for Problem 2 (the inner product), where the sum-of-products structure exhibits a high degree of geometric symmetry.

6.2 Operator Algebras and Hilbert Spaces — For Problem 2 (Inner Product)

Regarding Problem 2, the dot product of two vectors u and v does not necessarily have to be viewed in the algebraic form $\sum u_i v_i$.

Attack: We upgrade scalar quantities to linear operators acting on a Hilbert space. The expression $\langle u, v \rangle$ becomes an evaluation of the spectrum or trace of these operators. By employing spectral theory [8], we can assess the relative magnitude of two operator systems based on eigenvalue bounds, without needing to calculate each individual element precisely.

Quantum mechanics has employed this technique for decades: rather than precisely measuring the scalar state of a particle, one compares observable operators. This model provides an excellent fundamental mathematical framework for achieving a lower COST_ALL in establishing order. However, this approach addresses the problem on quantum hardware, rather than on traditional silicon hardware.

The pursuit of the space $\mathbb{S}$ is not merely a problem of computer algorithm optimization, but an effort to discover a new “semantics” for arithmetic expressions—one where order arises from the shape of the structure rather than from the result of a calculation.

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Index

adaptive-precision arithmetic, 4

classic processing workflow, 2

Classical Baseline, 3

COST_ALL

 computation cost, 3

 memory/representation cost, 3

 other overheads, 3

Hausdorff distance, 5

Hilbert space, 5

interval arithmetic, 4

Johnson–Lindenstrauss lemma, 4

Locality-Sensitive Hashing (LSH), 4

Main Problem 1 (General Case), 2

Main Problem 2 (Scalar Product Case), 3

Maximum Inner Product Search (MIPS), 4

operator algebras, 5

Requirement A1 (Comparability), 2

Requirement A2 (Structural Distance), 2

Requirement A3 (Total Ordering), 2

Richardson’s Theorem, 3

spectral theory, 5

tropical geometry, 4

Wasserstein distance, 5